Python -Learning

Notes by Kundan Kumar(KK)

Python Cheat

Base Types

integer, float, boolean, string, bytes

```
int 783  0 -192  0b010 0o642 0xF3
                 null  binary octal hexa
float 9.23 0.0  -1.7e-6
                ×10^-6
bool True False
str "One\nTwo"
    escaped new line
    'I\'m'
    escaped '
```

Multiline string: `"""X\tY\tZ 1\t2\t3"""` *escaped tab*

```
bytes b"toto\xfe\775"
      hexadecimal octal
```

immutables

Container Types

- **ordered sequences**, fast index access, repeatable values

```
list [1,5,9]   ["x",11,8.9]   ["mot"]   []
tuple (1,5,9)  11,"y",7.4     ("mot",)  ()
```

Non modifiable values (immutables) — *expression with just comas → `tuple`*

```
str bytes   (ordered sequences of chars / bytes)   "" b""
```

- **key containers**, no *a priori* order, fast key acces, each key is unique

```
dictionary  dict {"key":"value"}   dict(a=3,b=4,k="v")   {}
(key/value associations) {1:"one",3:"three",2:"two",3.14:"π"}
collection  set {"key1","key2"}   {1,9,3,0}   set()
```

keys=hashable values (base types, immutables…) — `frozenset` *immutable set* — *empty*

Identifiers

for variables, functions, modules, classes… names

`a…zA…Z_` followed by `a…zA…Z_0…9`

- diacritics allowed but should be avoided
- language keywords forbidden
- lower/UPPER case discrimination

```
☺ a toto x7 y_max BigOne
☹ 8y and for
```

Variables assignment

`=`

*assignment ⇔ **binding** of a name with a value*

1) evaluation of right side expression value

2) assignment in order with left side names

```
x=1.2+8+sin(y)
a=b=c=0      assignment to same value
y,z,r=9.2,-7.6,0   multiple assignments
a,b=b,a      values swap
a,*b=seq     unpacking of sequence in
*a,b=seq     item and list
x+=3         increment ⇔ x=x+3        and *= /= %= …
x-=2         decrement ⇔ x=x-2
x=None       « undefined » constant value
del x        remove name x
```

Conversions

`type(expression)`

```
int("15") → 15
int("3f",16) → 63        can specify integer number base in 2nd parameter
int(15.56) → 15          truncate decimal part
float("-11.24e8") → -1124000000.0
round(15.56,1)→15.6      rounding to 1 decimal (0 decimal → integer number)
bool(x)  False for null x, empty container x , None or False x ; True for other x
str(x)→ "…"   representation string of x for display (cf. formating on the back)
chr(64)→'@'  ord('@')→64   code ↔ char
repr(x)→ "…"  literal representation string of x
bytes([72,9,64]) → b'H\t@'
list("abc") → ['a','b','c']
dict([(3,"three"),(1,"one")]) → {1:'one',3:'three'}
set(["one","two"]) → {'one','two'}
separator str and sequence of str → assembled str
    ':'.join(['toto','12','pswd']) → 'toto:12:pswd'
str splitted on whitespaces → list of str
    "words with  spaces".split() → ['words','with','spaces']
str splitted on separator str → list of str
    "1,4,8,2".split(",") → ['1','4','8','2']
sequence of one type → list of another type (via comprehension list)
    [int(x) for x in ('1','29','-3')] → [1,29,-3]
```

Sequence Containers Indexing

for lists, tuples, strings, bytes…

negative index	-5	-4	-3	-2	-1
positive index	0	1	2	3	4
`lst=[10,`	`20,`	`30,`	`40,`	`50]`	
positive slice	0	1	2	3	4	5
negative slice	-5	-4	-3	-2	-1	

Items count

`len(lst)→5`

index from 0 (here from 0 to 4)

Individual access to **items** via `lst[index]`

```
lst[0]→10   ⇒ first one      lst[1]→20
lst[-1]→50  ⇒ last one       lst[-2]→40
```

On mutable sequences (`list`), remove with `del lst[3]` and modify with assignment `lst[4]=25`

Access to **sub-sequences** via `lst[start slice:end slice:step]`

```
lst[:-1]→[10,20,30,40]   lst[::-1]→[50,40,30,20,10]   lst[1:3]→[20,30]   lst[:3]→[10,20,30]
lst[1:-1]→[20,30,40]     lst[::-2]→[50,30,10]         lst[-3:-1]→[30,40] lst[3:]→[40,50]
lst[::2]→[10,30,50]      lst[:]→[10,20,30,40,50] shallow copy of sequence
```

Missing slice indication → from start / up to end.

On mutable sequences (`list`), remove with `del lst[3:5]` and modify with assignment `lst[1:4]=[15,25]`

Boolean Logic

Comparators: `< > <= >= == !=` (*boolean results*) ≤ ≥ = ≠

`a and b` logical and *both simulta-neously*

`a or b` logical or *one or other or both*

pitfall : `and` and `or` return value of `a` or of `b` (under shortcut evaluation).

⇒ ensure that `a` and `b` are booleans.

`not a` logical not

`True` `False` True and False constants

Statements Blocks

```
parent statement:
    statement block 1…
    ⋮
parent statement:
    statement block2…
    ⋮
next statement after block 1
```

indentation !

configure editor to insert 4 spaces in place of an indentation tab.

Modules/Names Imports

module `truc`⇔file `truc.py`

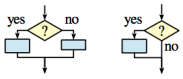
```
from monmod import nom1,nom2 as fct
                   →direct acces to names, renaming with as
import monmod →acces via monmod.nom1 …
```

*modules and packages searched in *python path* (cf `sys.path`)*

Conditional Statement

statement block executed only if a condition is true

```
if logical condition:
    statements block
```



Can go with several *elif*, *elif*… and only one final *else*. Only the block of first true condition is executed.

```
if age<=18:
    state="Kid"
elif age>65:
    state="Retired"
else:
    state="Active"
```

with a var `x`:

```
if bool(x)==True:  ⇔ if x:
if bool(x)==False: ⇔ if not x:
```

Maths

floating numbers… approximated values — *angles in radians*

Operators: `+ - * / // % **`

Priority `(…)` — × ÷ ↑ ↑ — a^b — integer ÷ — ÷ remainder

`@` → matrix × *python3.5+numpy*

```
(1+5.3)*2→12.6
abs(-3.2)→3.2
round(3.57,1)→3.6
pow(4,3)→64.0
```

usual priorities

```
from math import sin,pi…
sin(pi/4)→0.707…
cos(2*pi/3)→-0.4999…
sqrt(81)→9.0   √
log(e**2)→2.0
ceil(12.5)→13
floor(12.5)→12
```

modules `math`, `statistics`, `random`, `decimal`, `fractions`, `numpy`, *etc. (cf. doc)*

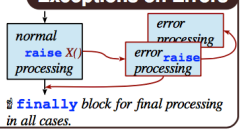
Exceptions on Errors

Signaling an error:

`raise` *ExcClass(…)*

Errors processing:

```
try:
    normal procesising block
except Exception as e:
    error processing block
```



`finally` *block for final processing in all cases.*

Conditional Loop Statement

statements block executed ***as long as*** *condition is true*

```
while logical condition:
    statements block
```

```
s = 0
i = 1
while i <= 100:
    s = s + i**2
    i = i + 1
print("sum:",s)
```

initializations ***before*** *the loop condition with a least one variable value (here* `i`*)*

make condition variable change !

beware of infinite loops!

Algo: $s = \sum_{i=1}^{i=100} i^2$

Loop Control

`break` *immediate exit*

`continue` *next iteration*

`else` *block for* ***normal*** *loop exit.*

Display

```
print("v=",3,"cm :",x,",",y+4)
```

items to display : literal values, variables, expressions

`print` options:

- `sep=" "` items separator, default space
- `end="\n"` end of print, default new line
- `file=sys.stdout` print to file, default standard output

Input

```
s = input("Instructions:")
```

`input` always returns a **string**, convert it to required type (cf. boxed *Conversions* on the other side).

Generic Operations on Containers

`len(c)` → items count

`min(c)` `max(c)` `sum(c)`

`sorted(c)` → `list` sorted *copy*

`val in c` → boolean, membership operator `in` (absence `not in`)

`enumerate(c)` → *iterator* on (index, value)

`zip(c1,c2…)` → *iterator* on tuples containing c_1 items at same index

`all(c)` → `True` if **all** `c` items evaluated to true, else `False`

`any(c)` → `True` if **at least one** item of `c` evaluated true, else `False`

Note: For dictionaries and sets, these operations use ***keys****.*

Specific to ***ordered sequences containers*** *(lists, tuples, strings, bytes…)*

`reversed(c)` → *inversed iterator* `c*5` → duplicate `c+c2` → concatenate

`c.index(val)` → *position* `c.count(val)` → *events count*

```
import copy
copy.copy(c) → shallow copy of container
copy.deepcopy(c) → deep copy of container
```

Operations on Lists

modify original list

`lst.append(val)`	add item at end
`lst.extend(seq)`	add sequence of items at end
`lst.insert(idx, val)`	insert item at index
`lst.remove(val)`	remove first item with value *val*
`lst.pop([idx])` → *value*	remove & return item at index *idx* (default last)
`lst.sort()` `lst.reverse()`	sort / reverse liste *in place*

Operations on Dictionaries

`d[key]=value` `d.clear()`

`d[key]` → *value* `del d[key]`

`d.update(d2)` { update/add associations

`d.keys()` `d.values()` `d.items()` } → *iterable views on keys/values/associations*

`d.pop(key[,default])` → *value*

`d.popitem()` → *(key,value)*

`d.get(key[,default])` → *value*

`d.setdefault(key[,default])` → *value*

Operations on Sets

Operators:

- `|` → union (vertical bar char)
- `&` → intersection
- `-` `^` → différence/symetric diff.
- `<` `<=` `>` `>=` → inclusion relations

Operators also exist as methods.

`s.update(s2)` `s.copy()`

`s.add(key)` `s.remove(key)`

`s.discard(key)` `s.clear()`

`s.pop()`

Files

storing data on disk, and reading it back

```
f = open("file.txt","w",encoding="utf8")
```

file **variable** for operations

name of file on disk (+path…)

opening **mode**
- `'r'` read
- `'w'` write
- `'a'` append
- …`'+'` `'x'` `'b'` `'t'`

encoding of chars for *text files*: `utf8` `ascii` `latin1` …

cf. modules `os`, `os.path` and `pathlib`

writing

`f.write("coucou")`

`f.writelines(list of lines)`

reading

read empty string if end of file

`f.read([n])` → *next chars* *if n not specified, read up to end !*

`f.readlines([n])` → `list` *of next lines*

`f.readline()` → *next line*

text mode `t` *by default (read/write* `str`*), possible binary mode* `b` *(read/write* `bytes`*). Convert from/to required type !*

`f.close()` dont forget to **close the file** after use !

`f.flush()` write cache `f.truncate([taille])` resize

reading/writing progress sequentially in the file, modifiable with:

`f.tell()` → *position* `f.seek(position[,origin])`

Very common: opening with a guarded block (automatic closing) and reading loop on lines of a text file:

```
with open(…) as f:
    for line in f :
        # processing of line
```

Iterative Loop Statement

statements block executed ***for each*** *item of a container or iterator*

```
for var in sequence:
    statements block
```

Go over sequence's **values**

```
s = "Some text"
cnt = 0
for c in s:
    if c == "e":
        cnt = cnt + 1
print("found",cnt,"'e'")
```

initializations ***before*** *the loop*

loop variable, ***assignment managed by*** `for` ***statement***

Algo: count number of e *in the string.*

loop on dict/set ⇔ loop on keys sequences

use *slices* to loop on a subset of a sequence

Go over sequence's **index**

- modify item at index
- access items around index (before / after)

```
lst = [11,18,9,12,23,4,17]
lost = []
for idx in range(len(lst)):
    val = lst[idx]
    if val > 15:
        lost.append(val)
        lst[idx] = 15
print("modif:",lst,"-lost:",lost)
```

Algo: limit values greater than 15, memorizing of lost values.

good habit : don't modify loop variable

Go simultaneously on sequence's **index** and **values**:

```
for idx,val in enumerate(lst):
```

Integers Sequences

`range([start,] end [,step])`

start default 0, *fin* not included in sequence, *pas* signed default 1

`range(5)` → 0 1 2 3 4 `range(2,12,3)` → 2 5 8 11

`range(3,8)` → 3 4 5 6 7 `range(20,5,-5)` → 20 15 10

`range(len(seq))` → *sequence of index of values in seq*

range provides an immutable sequence of int constructed as needed

Function Definition

function name (identifier)

named parameters

```
def fct(x,y,z):
    """documentation"""
    # statements block, res computation, etc.
    return res
```

result value of the call, if no computed result to return: `return None`

parameters and all variables of this block exist only *in* the block and *during* the function call (think of a "black box")

Advanced: `def fct(x,y,z,*args,a=3,b=5,**kwargs):`

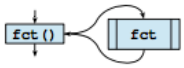
**args variable positional arguments (→* `tuple`*), default values, **kwargs variable named arguments (→* `dict`*)*

Function Call

```
r = fct(3,i+2,2*i)
```

storage/use of returned value

one argument per parameter



this is the use of function name *with parenthesis* which does the call

Advanced:
**sequence*
***dict*

Operations on Strings

`s.startswith(prefix[,start[,end]])`

`s.endswith(suffix[,start[,end]])` `s.strip([chars])`

`s.count(sub[,start[,end]])` `s.partition(sep)` → *(before,sep,after)*

`s.index(sub[,start[,end]])` `s.find(sub[,start[,end]])`

`s.is…()` *tests on chars categories (ex.* `s.isalpha()`*)*

`s.upper()` `s.lower()` `s.title()` `s.swapcase()`

`s.casefold()` `s.capitalize()` `s.center([width,fill])`

`s.ljust([width,fill])` `s.rjust([width,fill])` `s.zfill([width])`

`s.encode(encoding)` `s.split([sep])` `s.join(seq)`

Formating

formating directives — values to format

```
"modele{} {} {}".format(x,y,r) → str
```

`"{selection : formating!conversion}"`

Selection :

`2`
`nom`
`0.nom`
`4[key]`
`0[2]`

Examples:

```
"{:+2.3f}".format(45.72793)
→'+45.728'
"{1:>10s}".format(8,"toto")
→'      toto'
"{x!r}".format(x="I'm")
→'"I\'m"'
```

Formating :

fill char alignment sign mini width.precision~maxwidth type

`< > ^ =` `+ - space` `0` at start for filling with 0

integer: `b` binary, `c` char, `d` decimal (default), `o` octal, `x` or `X` hexa…

float: `e` or `E` exponential, `f` or `F` fixed point, `g` or `G` appropriate (default),

string: `s` … `%` percent

Conversion : `s` (readable texte) or `r` (literal representation)

Variable Assignment

```
integer = 1
string = "string"
unicode_string = u"unicode string"
mutli_line_string = """ multi-line
        string
        """
tuple = (element1, element2, element3, ...)
list = [ element1, element2, element3, ... ]
dictionary = { key1 : value1, key2 : value2, ... }
dictionary[key] = value
class_instance = ClassName(init_args)
```

Frequently Used Built-in Types

True	**False**	**None**
str	**unicode**	**int**
float	**list**	**dict**

Other than **True**, **False** and **None**, these can also be used as functions to explicitly cast a value to that type

Functions

```
def function_name(arg1, arg2,
                  keyword1=val1, keyword2=val2, ...):
    <function body>
    return return_value
```

e.g.

```
def my_function(x, y, z=0):        my_function(1, 2) → 3
    sum = x + y + z                my_function(1, 2, 3) → 6
    return sum                     my_function(1, 2, y=4) → 7
```

Classes

```
class ClassName(SuperClass):
    class_variable = static_value
    def __init__(self, value1, <...>):
        self.instance_variable1 = value1
        self.instance_function()
    def instance_function(self, arg1, <...>):
        <function body>
        return return_value
```

e.g.

```
class MyClass(object):             MyClass.offset → 1
    offset = 1
    def __init__(self, value):     c = MyClass(2)
        self.value = value         c.value → 2
    def get_offset_value(self):    c.get_offset_value() → 3
        return MyClass.offset +
               self.value
```

Imports

```
import module
from module import class, function, variable
```

Frequently Used String Manipulations

string1 + *string1*	"str" + "ing" → "string"
"%s%s" % (*string1*, *string2*)	"%s%s" % ("s", "g") → "sg"
string.split("*delim*", *limit*)	"s/g".split("/") → ["s", "g"]
string.strip()	" string ".strip() → "string"
string.startswith("*prefix*")	"str".startswith("s") → True
substring in *string*	"str" in "string" → True

print *string*

List Comprehension

[*value* **for** *value* **in** *list* **if** *condition*]

e.g.

[x for x in [1,2,3,4,5,6,7,8,9] if x % 2 == 0] → [2,4,6,8]

Accessing Variable Values

value = *dictionary*[*key*]	
value = *dictionary*.get(*key*, *default_value*)	
value = *list*[*index*]	e.g. [5,6,7][2] → 7
value = *string*[*start*:*end*]	e.g "string"[0:3] → "str"
value = *list*[*start*:*end*]	e.g. [1,2,3][1:2] → [2]
value = *ClassName.class_variable*	
value = *class_instance.instance_variable*	
value = *class_instance.function(args)*	

Comparisons

value1 == *value2*	"str" == "str" → **True**
value1 != *value2*	"str" != "str" → **False**
value1 < *value2*	1 < 2 → **True**
value1 <= *value2*	2 <= 2 → **True**
value1 > *value2*	2 > 3 → **False**
value1 >= *value2*	3 >= 3 → **True**
value **is [not] None**	
value **in** *list*	1 in [2,3,4] → **False**

isinstance(*class_instance*, *ClassName*)

Basic Arithmetic

i = a + b	i = a - b
i = a / b	i = a * b
i = a % b	e.g. 11 % 3 → 2

Comments

```
"""                          # Line Comment
    Multi-line comment
"""
```

Control Flow

```
if conditional:                  if i == 7:
    <body>                           print "seven"
elif conditional:          e.g.  elif i == 8:
    <body>                           print "eight"
else:                            else:
    <body>                           print str(i)
for value in list:               for i in [1, 2, 3, 4]:
    <body>                 e.g.      if i == 2: continue
    continue                         if i == 3: break
    break                            print i
while conditional:               while True:
    <body>                 e.g.      print "infinity"
    continue
    break
```

Exceptions

```
try:                             try:
    <body>                           database.update()
    raise Exception()          e.g. except Exception as e:
except Exception as e:               log.error(e.msg)
    <exception handling>             database.abort()
finally:                         finally:
    <clean-up>                       database.commit()
```

File & Path Manipulation

```
import os    # import the os module first
os.path.join(path_segment1, path_segment2, ...)
os.path.exists(path)
os.listdir(directory_path)
os.remove(file_path)
os.rmdir(directory_path)
file = open(path, "rw")
file.read()
string.write("string")
```

```
A = LOAD 'Anuj/2014/00' USING PigStorage(',') AS
(roadtmpc:double,srad:double,dwpc:double,ptype:integer,tmpc:double,pcpn:double,smps:double,wawa:chararray,vs
by:double,snwd:double,gid:integer,drct:integer);
B = filter A by gid > 0 ;
C = foreach B generate gid,pcpn ;
D = group C by gid;
E = foreach D generate group,SUM(C.pcpn) as sum,AVG(C.pcpn);

F = filter F by sum > 0;

STORE E INTO 'Anuj/result' USING org.apache.pig.piggybank.storage.CSVExcelStorage(',');


E = foreach D generate group,SUM(C.pcpn) as sum,AVG(C.pcpn)as avg;

F = filter E by avg > 0;

A = LOAD 'Anuj/Combined.csv' using PigStorage(',');

 B = limit A 100000;

C = store B into 'Anuj/Combined_Sample1' USING org.apache.pig.piggybank.storage.CSVExcelStorage(',');
```

https://talendexpert.com/coursera-assignment-advanced-join-in-spark-submit-this-file/